CEB-ViTAL: Coverage-Aware, Energy-Balanced Clustering for Lifetime Extension in Wireless Sensor Networks

Ali Hamza and Javed Iqbal

***Abstract*—ViTAL is a recent three-phase lifetime-extension approach for wireless sensor networks (WSNs). It combines harmony-search deployment for coverage, a connected dominating set (CDS), and a four-factor cluster-head (CH) score. We show that its CH score is almost insensitive to residual energy, favours CHs far from the sink, and concentrates the CH role on a small CDS backbone. We also note that coverage, although optimised at deployment, is ignored during operation.**

We propose CEB-ViTAL, which keeps ViTAL's deployment phase and adds three mechanisms: (i) a coverage-preserving, energy-weighted sleep scheduler; (ii) a normalised four-factor score with a dynamic energy weight, used to elect the analytically optimal number of spatially separated CHs; (iii) energy-aware multi-hop forwarding from CHs to the sink.

All protocols were evaluated in one shared simulator over 6,900 runs on 30 held-out random networks, six scenarios and 100–500 nodes. Hypotheses and constants were pre-registered.

In the base setting of ViTAL (100 m field, corner sink), CEB-ViTAL extends the 90% coverage lifetime by +21% to +130% over the strongest baseline. It extends first-node death by +40% to +184% over LEACH and +224% to +2519% over ViTAL, and the coverage integral by +15% to +148%. Every pre-registered test is significant (Holm-corrected $p \leq 6.1 \times 10^{-6}$). Readings delivered differ by -4% to +1% from always-on protocols. Gains shrink in sparse deployments. In the base setting they remain positive even when sleeping is charged as much as transmitting.

***Index Terms*—Wireless sensor networks, clustering, LEACH, coverage preservation, sleep scheduling, energy balance, network lifetime.**

I. Introduction

WIRELESS sensor networks (WSNs) support environmental monitoring, health care, smart homes and industrial IoT [1]. Their nodes run on batteries that are rarely replaced, so energy efficiency decides how long a deployment can serve its purpose.

Hierarchical clustering, introduced by LEACH [2], [3], rotates the cluster-head (CH) role so that the high cost of long-range transmission is shared. Many successors refine the CH election with energy, distance and topology information [4]–[7].

ViTAL [8] argues that deployment, topology and radio conditions must be treated together. It has three phases:

- a harmony-search (HSA) deployment that maximises coverage [9], [10];
- a connected-dominating-set (CDS) backbone [11];
- a four-factor (4F) CH score built from residual energy, neighbour count, distance to the sink and signal-to-noise ratio (SNR).

ViTAL reports large gains over LEACH and two of its variants, and names dynamic weighting and scheduling as future work.

This paper continues that line of work. Re-implementing ViTAL from its description, we found three weaknesses in how its CH score is normalised and applied. We also note a missed opportunity: the coverage that phase 1 optimises is never used during operation, so every node senses and reports every round even where its area is already covered.

Contributions.

1) An analysis of ViTAL's CH selection. Its energy term scales as $1/N$, its distance term favours far nodes, and its CDS restriction concentrates load (Section IV).
2) CEB-ViTAL. The protocol couples ViTAL's coverage-driven deployment with coverage-preserving, energy-weighted sleep scheduling and a corrected, energy-balanced clustering phase (Section V).
3) A pre-registered evaluation on held-out random networks against LEACH, DEEC, ViTAL and our intermediate design EB-ViTAL. It covers six scenarios, includes ablations and a sleep-energy robustness check, and uses paired non-parametric tests with multiplicity control (Sections VI–VII).

II. Related Work

[Authors: extend this section with a systematic 2023–2026 literature search on coverage-preserving clustering and duty cycling before submission.]

Clustering protocols.

- LEACH [2] elects CHs at random, with probability p and an epoch-based rotation.
- LEACH-C [3] centralises the election at the sink and derives the optimal number of clusters.
- SEP [4] and DEEC [6] weight the CH probability by initial or residual energy relative to the network average.
- HEED [5] combines residual energy with a communication cost and a competition range that spreads CHs.
- PEGASIS [12] replaces clusters with a chain.
- Multi-hop LEACH variants such as EEM-LEACH [13] relay CH traffic to avoid long single-hop links.

Coverage-preserving scheduling. When nodes are deployed densely, their sensing areas overlap, and redundant nodes can sleep without losing coverage.

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- Slijepcevic and Potkonjak [14] and Cardei and Du [15] organise nodes into covering sets that are activated in turn.
- Tian and Georganas [16] let each node decide locally whether its area is already covered by neighbours.
- CCP [17] and OGDC [18] jointly maintain coverage and connectivity.
- Randomised independent sleeping [19] provides analytical coverage guarantees.
- Soro and Heinzelman [20] make CH selection itself coverage-aware.
- Greedy weighted set cover [21] is the standard approximation for choosing a covering subset.

ViTAL [8]. It integrates HSA deployment, a CDS and 4F CH selection. The present work keeps its deployment phase, replaces its CDS-restricted 4F election, and adds coverage-preserving scheduling. Combining coverage scheduling with corrected, energy-balanced clustering on top of a coverage-optimised deployment is the contribution claimed here.

III. System Model

A. Network and energy model

N static sensors are deployed in an $M \times M$ field. The sink is either static or follows a random-waypoint path. Every node starts with energy E_0. In each round, every *sensing* node produces one L-bit reading. Members send their readings to a CH, which aggregates them and forwards one packet towards the sink.

We use the first-order radio model [3]:

$$E_{\mathrm{TX}}(L,d) = \begin{cases} LE_{\mathrm{elec}} + L\varepsilon_{\mathrm{fs}} d^2, & d < d_0, \\ LE_{\mathrm{elec}} + L\varepsilon_{\mathrm{mp}} d^4, & d \geq d_0, \end{cases} \tag{1}$$

with $E_{\mathrm{RX}}(L) = LE_{\mathrm{elec}}$ and $d_0 = \sqrt{\varepsilon_{\mathrm{fs}}/\varepsilon_{\mathrm{mp}}}$. Aggregation costs E_{DA} per bit per signal. Table I lists all parameters.

B. Coverage model and metrics

As in ViTAL, the field is divided into 1 m demand cells, and node i covers cell c if $\|\mathbf{p}_c - \mathbf{x}_i\| \leq R_s$. Let $\mathcal{A}(r)$ be the set of sensing nodes in round r and $\mathcal{C}_i$ the cells covered by node i. The covered fraction is

$$C(r) = \frac{1}{M^2} \Big| \bigcup_{i \in \mathcal{A}(r)} \mathcal{C}_i \Big|, \tag{2}$$

and C_0 is the value of $C(0)$ with all nodes active. The primary metric is the *coverage lifetime*

$$T_{90} = \min\{r : C(r) < 0.9\, C_0\}. \tag{3}$$

We also report:

- the coverage integral $\sum_r C(r)$, in coverage-rounds;
- first, half and last node death (FND, HND, LND);
- the number of readings delivered to the sink.

TABLE I
Simulation Parameters

Parameter	Value
Field size M	100 m (S1, S2, S5, S6), 200 m (S3, S4)
Nodes N	100, 200, 300, 400, 500
Initial energy E_0	0.5 J
E_{elec}, E_{DA}	50 nJ/bit, 5 nJ/bit/signal
$\varepsilon_{\mathrm{fs}}$, $\varepsilon_{\mathrm{mp}}$	10 pJ/bit/m^2, 0.0013 pJ/bit/m^4
Packet size L	800 bits (4000 bits in S6)
Sensing / neighbour range	R_s = 5 m / 15 m
LEACH CH probability p	0.1
SNR model	P_t 0 dBm, PL_0 40 dB, n 3, σ 4 dB, noise -100 dBm
HSA (phase 1)	HMS 10, HMCR 0.9, PAR 0.3, BW 5, 2000 it.
CovSched	$\tau = 0.95$, $r_s = 10$, $L_{\mathrm{ctrl}} = 200$ bits
Seeds	30 held-out networks (1000–1029)

IV. Analysis of ViTAL's Cluster-Head Selection

ViTAL elects node n as CH if a uniform random number is below $\mathrm{NV}_n \cdot T(n)$. Here $T(n)$ is the LEACH threshold, and the normalisation vector is

$$\mathrm{NV}_n = w_1 \frac{E_{\mathrm{res}}}{E_{\mathrm{res}}^{\mathrm{total}}} + w_2 \frac{\mathrm{Neigh}}{N} + w_3 \frac{\mathrm{Dist}}{\overline{\mathrm{Dist}}} + w_4 \frac{\mathrm{SNR}}{\overline{\mathrm{SNR}}}, \tag{4}$$

with $\sum w_k = 1$ [8]. We identify four issues.

G1, near energy-blindness. The energy term divides a node's energy by the *network total*, so it is approximately $1/n_{\mathrm{alive}}$. The distance and SNR terms, in contrast, are approximately 1. At $N = 100$ with equal weights, the energy term contributes about 0.0025, against about 0.25 for each of the other terms. Residual energy therefore barely influences the election. When a deterministic election uses (4) unchanged, the same nodes are chosen repeatedly and first-node death falls by 93–97% compared with our corrected score.

G2, inverted distance preference. $\mathrm{Dist}/\overline{\mathrm{Dist}}$ increases with distance, so the score favours CHs *far* from the sink. With the sink in a corner, as in ViTAL's set-up, such CHs pay the d^4 amplifier cost.

G3, CDS bottleneck. Restricting candidates to the CDS concentrates the CH role on a backbone of about 28 nodes, whatever N is. In our simulations ViTAL has the shortest FND of all protocols, and its FND falls as N grows (Table II).

G4, no coverage awareness. Coverage is maximised at deployment and then ignored. In dense deployments many nodes are redundant, yet all of them spend energy every round.

V. Proposed Method: CEB-ViTAL

Fig. 1 summarises the three phases. Like LEACH-C and ViTAL, CEB-ViTAL is coordinated by the sink, which knows node positions and residual energies.

A. Phase 1: coverage-driven deployment

Phase 1 is unchanged from ViTAL [8], [10]. Harmony search places the N nodes to maximise the coverage ratio multiplied by the normalised minimum inter-node distance, using the parameters in Table I.

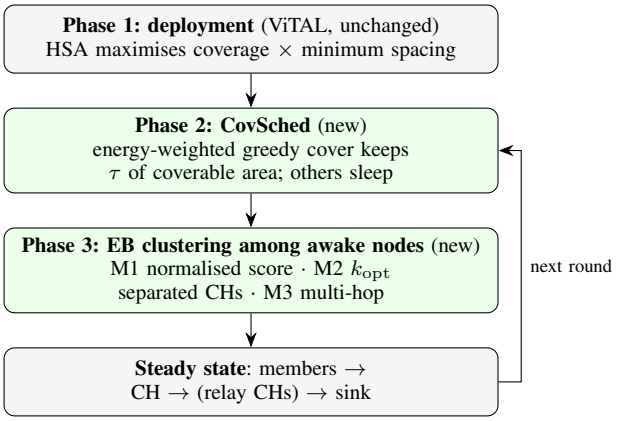


Fig. 1. CEB-ViTAL. Phase 1 is inherited from ViTAL [8]; phases 2 and 3 are the contributions of this work. The active set is recomputed every r_s rounds and whenever an active node dies.

Require: alive set $\mathcal{L}$, energies E, cell sets $\mathcal{C}_i$, target τ
$\mathcal{U} \leftarrow \bigcup_{i\in\mathcal{L}} \mathcal{C}_i$; $\mathcal{A} \leftarrow \emptyset$; $\mathcal{K} \leftarrow \emptyset$
while $|\mathcal{K}| < \tau|\mathcal{U}|$ **do**
  $i^\star \leftarrow \arg\max_{i\in\mathcal{L}\setminus\mathcal{A}} |\mathcal{C}_i \setminus \mathcal{K}| \cdot E_i / \max_{j\in\mathcal{L}} E_j$
  if $|\mathcal{C}_{i^\star} \setminus \mathcal{K}| = 0$ **then break**
  end if
  $\mathcal{A} \leftarrow \mathcal{A} \cup \{i^\star\}$; $\mathcal{K} \leftarrow \mathcal{K} \cup \mathcal{C}_{i^\star}$
end while
broadcast $\mathcal{A}$; nodes in $\mathcal{L} \setminus \mathcal{A}$ sleep

Algorithm 1: CovSched (run at the sink)

B. Phase 2: coverage-preserving sleep scheduling (CovSched)

Let $\mathcal{U}$ be the set of cells coverable by all alive nodes. Every r_s rounds, and whenever an active node dies, the sink computes an active set $\mathcal{A}$ with an energy-weighted greedy set cover (Algorithm 1) [21]. Each step selects

$$i^\star = \arg\max_{i\notin\mathcal{A}} \left|\mathcal{C}_i \setminus \bigcup_{j\in\mathcal{A}} \mathcal{C}_j\right| \cdot \frac{E_i}{\max_j E_j} \tag{5}$$

until $\tau|\mathcal{U}|$ cells are covered.

Sleeping nodes neither sense nor transmit. Every alive node pays $L_{\mathrm{ctrl}}E_{\mathrm{elec}}$ per reschedule to receive the schedule. Weighting by residual energy rotates the sensing duty towards nodes with more energy. The target $\tau < 1$ prevents a single node with a unique cell from being kept awake permanently.

C. Phase 3: energy-balanced clustering among awake nodes

M1, normalised score with dynamic energy weight. Each factor is divided by its maximum over the awake nodes, so that all four lie in $[0, 1]$, and the distance term is inverted so that nodes closer to the sink score higher:

$$s_i = w_1(t)\frac{E_i}{\max E} + w(t)\Big[\frac{\mathrm{Neigh}_i}{\max \mathrm{Neigh}} + \Big(1 - \frac{\mathrm{Dist}_i}{\max \mathrm{Dist}}\Big) + \frac{\mathrm{SNR}_i}{\max \mathrm{SNR}}\Big]. \tag{6}$$

The weights are $w_1(t) = 0.25 + 0.5\,(1 - \bar{E}(t)/E_0)$ and $w(t) = (1 - w_1(t))/3$. The energy weight therefore grows from 0.25 to 0.75 as the network drains, which realises the dynamic weighting proposed as future work in [8].

M2, optimal number of separated CHs. The sink elects

$$k = \mathrm{round}\Big(\sqrt{\tfrac{n}{2\pi}}\sqrt{\tfrac{\varepsilon_{\mathrm{fs}}}{\varepsilon_{\mathrm{mp}}}}\frac{M}{\bar{d}^2}\Big) \tag{7}$$

CHs [3], where n is the number of awake nodes and $\bar{d}$ their mean distance to the sink. Candidates are scanned in descending s_i. A candidate is accepted if it lies at least $r_{\mathrm{sep}} = \sqrt{M^2/(\pi k)}$ from every CH already chosen. This deterministic, spatially spread election replaces ViTAL's random threshold and its CDS restriction (G3).

M3, energy-aware multi-hop forwarding. The sink computes a shortest-path tree over the CHs and itself. The cost of a link from CH i to CH j is $E_{\mathrm{TX}}(L, d_{ij}) + LE_{\mathrm{elec}}\,\bar{E}/E_j$; a direct link to the sink costs $E_{\mathrm{TX}}(L, d_{i,\mathrm{sink}})$. Relays forward packets without re-aggregation. Members join the nearest CH, or the sink if it is closer.

Complexity. One schedule costs $O(|\mathcal{A}|\cdot\mathrm{nnz})$, where $\mathrm{nnz} \approx 78N$ is the number of node–cell incidences. One election costs $O(N \log N + kN)$, and routing costs $O(k^2)$ on $k+1$ vertices.

VI. Evaluation Methodology

Simulator. All protocols run in one Python round-based simulator. They share the radio model, the deployments and the random streams, so a given seed yields the same network and the same sink trajectory for every protocol.

Compared protocols.

- LEACH [2], on random and on HSA deployment;
- DEEC [6];
- ViTAL [8], re-implemented from its description (see Section VIII);
- EB-ViTAL, which is phase 3 alone with every node always on;
- CEB-ViTAL.

Ablations.

- LEACH clustering with CovSched;
- EB-ViTAL with *random* sleepers, choosing the same number of active nodes as CovSched;
- CEB-ViTAL without M3.

Scenarios.

- S1: ViTAL's setting, a 100 m field with the sink at $(5, 5)$;
- S2: centre sink;
- S3 and S4: 200 m fields with corner and centre sinks;
- S5: a mobile sink (random waypoint, 1–10 m/s, 10 s pause);
- S6: as S1 with 4000-bit packets.

Pre-registration and held-out evaluation. The design, all constants, the scenarios and three hypotheses were fixed before testing:

- H1: CEB-ViTAL has a longer T_{90} than each comparator in S1;
- H2: it has a larger coverage integral;
- H3: it has a longer FND.

The comparators are LEACH (HSA), DEEC, ViTAL and EB-ViTAL. Each hypothesis is tested with a one-sided paired Wilcoxon signed-rank test [22] at each N, with Holm correction [23] over its 20 comparisons. All reported runs use 30 network seeds that were never used during development.

Fairness check. Before testing, the simulator reproduced 7,620 earlier runs bit-for-bit.

Uncertainty. Values are means with 95% confidence intervals.

VII. Results

A. Base setting (S1)

Table II and Fig. 2 report S1, and Table III reports the hypothesis tests. All three hypotheses hold in all 20 comparisons.

CEB-ViTAL's coverage lifetime grows with density because more nodes are redundant. At $N = 300$ it reaches 10686 rounds, against 5758 for the strongest always-on protocol.

Fig. 3 shows the mechanism. CEB-ViTAL deliberately runs at about $\tau = 95\%$ of the always-on coverage and keeps that level roughly twice as long, while always-on protocols lose coverage once nodes begin to die.

Fig. 4 shows one test network ($N = 300$) at round 5500:

- LEACH has 230 nodes alive and covers 77% of the field.
- ViTAL has 163 alive, covers 67%, and elects no CH in that round, so every node sends directly to the sink.
- EB-ViTAL keeps all 300 nodes alive, but they are nearly drained, with a mean residual energy of 0.03 J.
- CEB-ViTAL also keeps all 300 nodes alive. Only 144 of them sense, they cover 81% of the field, and the mean residual energy is 0.27 J.

Fig. 5 shows why:

- With about 148 of 300 nodes sensing per round, CEB-ViTAL's mean residual energy falls roughly half as fast as that of the always-on protocols.
- The energy imbalance of the always-on protocols rises sharply once their first nodes die. CEB-ViTAL's imbalance stays low until the network approaches exhaustion.

Two costs must be noted:

- Readings delivered differ by -4% to $+1\%$ from the always-on protocols, because sleeping nodes do not report. The sensed area is kept at τ of what all alive nodes could cover.
- LND is not uniformly better. At $N = 100$ it is 7294 rounds, against 8954 for LEACH (HSA) and 11376 for ViTAL. At $N = 500$ it exceeds every always-on protocol (16415 vs. at most 11696).

B. Other scenarios

Table IV compares CEB-ViTAL with the strongest always-on protocol in each scenario. Fig. 6 summarises the T_{90} gain, which ranges from +2% to +140%, and Fig. 7 shows the 200 m corner-sink field.

Gains depend on how much redundant coverage the deployment offers. In S3 the T_{90} gain is only +2% to +17%, because 100 nodes cover just 19% of a 200 m field. In S1, by comparison, 100 nodes cover 56% and 500 nodes cover 92%.

The mobile sink (S5) and larger packets (S6) reproduce the S1 pattern.

C. Ablation

Table V isolates each component.

- **Scheduling vs. clustering.** Both phases matter. In the 100 m field, removing CovSched costs more than replacing the clustering for $N \geq 200$; at $N = 100$ the two are comparable. In the sparse 200 m field (S3), the EB clustering contributes more than the scheduler.
- **Node selection.** Choosing the same number of sleepers at random drops coverage below $0.9C_0$ within the first rounds in all but the sparsest settings, so selection matters as much as the number of active nodes.
- **Clustering.** Replacing EB clustering with LEACH under the same scheduler lowers T_{90}; CEB-ViTAL is +23% to +61% better in S1.
- **Multi-hop forwarding (M3).** It has no measurable effect in the 100 m fields, where relays are rarely cheaper than direct links. In S3 its effect is 0% to +3%.

D. Robustness to sleep energy

The radio model treats sleeping as free. We therefore added a check that was *not* pre-registered: every sleeping node is charged fLE_{elec} per round. With $f = 1$, sleeping costs as much as the electronics energy of sending one packet. This penalises only the scheduled protocols.

Table VI shows the results.

- In S1, CEB-ViTAL keeps a positive T_{90} advantage even at $f = 1$ (+10% to +56%).
- In the sparse S3 field, the T_{90} advantage disappears at $f = 1$ (-3% to +6%), although FND remains much longer.

VIII. Discussion and Threats to Validity

Coverage level. CEB-ViTAL trades a fixed share of instantaneous coverage ($1 - \tau = 5\%$) for a much longer coverage lifetime. Applications that need every coverable cell at every instant should raise τ or use an always-on protocol.

ViTAL baseline. ViTAL is our re-implementation from its paper. Several of its parameters are not published, including E_{elec}, p, the weights w_1–w_4 and the SNR model. Our absolute ViTAL numbers therefore differ from those reported in [8], and comparisons with ViTAL should be read with that caveat. LEACH and DEEC are fully specified.

Model scope. The study uses the first-order radio model, with no packet loss, MAC contention or idle listening, and data links are not limited by radio range, as in LEACH-type studies. The sleep-energy check bounds one of these simplifications.

Centralised control. The scheduler and the election run at the sink, as in LEACH-C and ViTAL.

Constants. The constants τ, r_s, L_{ctrl} and the M1 weight schedule were fixed before testing and not tuned. Other values may perform better or worse.

IX. Conclusion

Coupling ViTAL's coverage-driven deployment with coverage-preserving, energy-weighted sleep scheduling and corrected, energy-balanced clustering substantially

TABLE II
SCENARIO S1 (100 M FIELD, SINK AT (5,5), 800-BIT PACKETS). MEAN ± 95% CI OVER 30 HELD-OUT NETWORKS.

	T_{90} (rounds)			FND (rounds)		
Protocol	N=100	N=300	N=500	N=100	N=300	N=500
LEACH (random)	5170±27	5386±18	5505±16	4825±46	4665±46	4549±53
LEACH (HSA)	5230±21	5508±18	5652±13	4931±39	4750±42	4650±37
DEEC	5502±18	5639±11	5674±10	5378±22	5330±24	5257±17
ViTAL	3048±60	4529±86	5193±61	2127±38	818±20	504±17
EB-ViTAL	5716±17	5758±10	5767±6	5676±37	5667±33	5650±24
LEACH + CovSched	5632±22	6645±54	8411±148	5621±22	6621±55	8324±148
CEB-ViTAL	**6900±58**	**10686±55**	**13290±181**	**6889±59**	**10642±57**	**13195±180**

	LND (rounds)		Coverage integral		Readings ($\times 10^3$)	
Protocol	N=100	N=500	N=100	N=500	N=100	N=500
LEACH (random)	10087±543	11187±324	3075±37	6025±11	586.1±2.8	2944.6±2.9
LEACH (HSA)	8954±330	9272±368	3346±35	5814±21	584.8±1.3	2944.5±2.4
DEEC	7447±375	7949±448	3287±35	5560±20	580.3±1.1	2948.5±2.2
ViTAL	11376±289	11696±216	3375±37	6695±37	565.5±3.5	2816.3±10.5
EB-ViTAL	6039±368	6760±459	3240±38	5426±25	575.0±1.5	2919.3±2.8
LEACH + CovSched	9226±324	18222±442	3502±27	10744±69	573.8±1.3	2858.7±3.1
CEB-ViTAL	**7294±309**	**16415±193**	**3723±24**	**13440±62**	**566.8±1.7**	**2820.0±2.9**

TABLE III
PRE-REGISTERED HYPOTHESES IN S1. ONE-SIDED PAIRED WILCOXON TEST WITH HOLM CORRECTION OVER 20 TESTS PER HYPOTHESIS.

Hyp.	Comparator	Gain (N=100–500)	Max. Holm p	Sig.
H1	LEACH (HSA)	+32% to +135%	5.2×10^{-6}	5/5
	DEEC	+25% to +134%	5.2×10^{-6}	5/5
	ViTAL	+126% to +156%	5.2×10^{-6}	5/5
	EB-ViTAL	+21% to +130%	5.2×10^{-6}	5/5
H2	LEACH (HSA)	+11% to +131%	1.9×10^{-8}	5/5
	DEEC	+13% to +142%	1.9×10^{-8}	5/5
	ViTAL	+10% to +101%	1.9×10^{-8}	5/5
	EB-ViTAL	+15% to +148%	1.9×10^{-8}	5/5
H3	LEACH (HSA)	+40% to +184%	6.1×10^{-6}	5/5
	DEEC	+28% to +151%	6.1×10^{-6}	5/5
	ViTAL	+224% to +2519%	6.1×10^{-6}	5/5
	EB-ViTAL	+21% to +134%	6.1×10^{-6}	5/5

extends coverage lifetime in dense networks, with only a small change in delivered readings. Future work includes distributed scheduling, validation against the original ViTAL implementation, packet-level simulation and a hardware testbed.

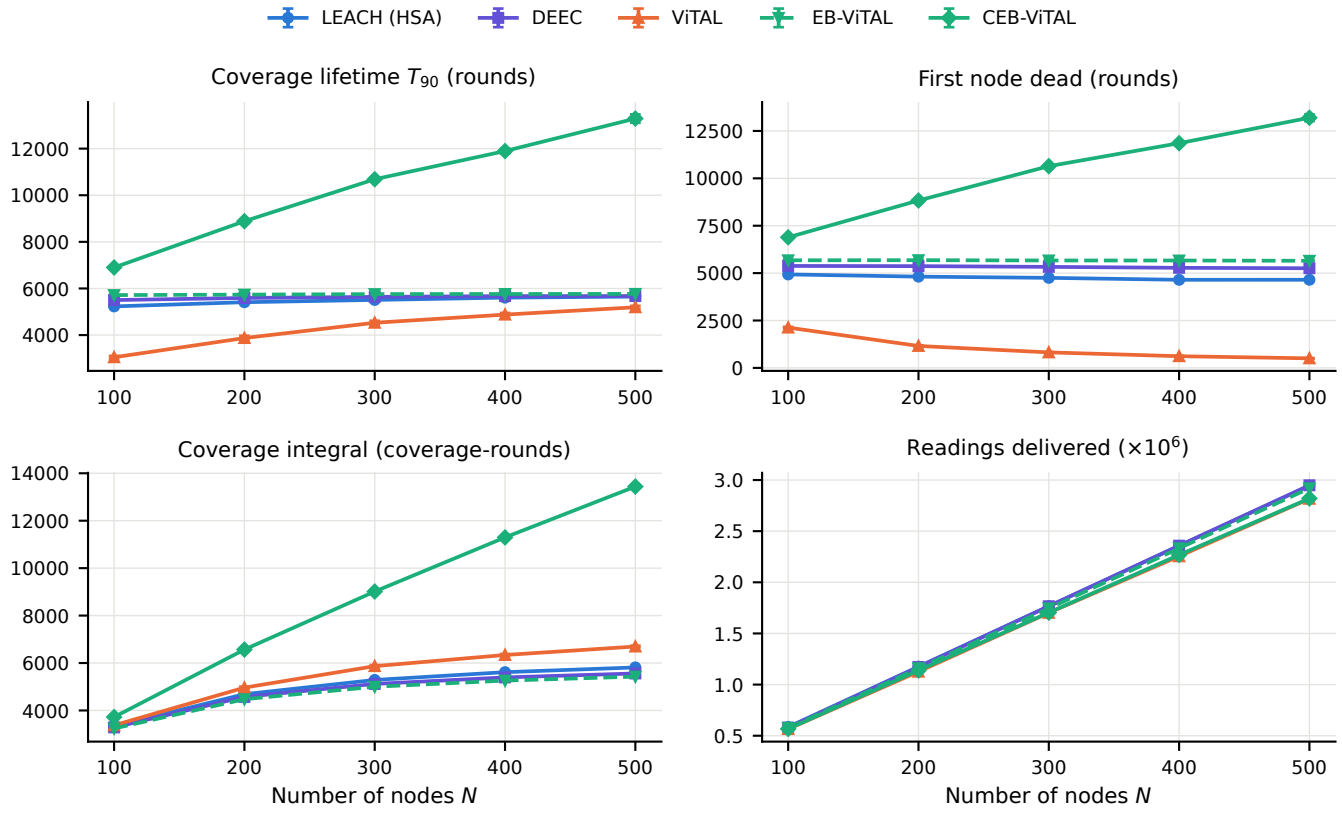


Fig. 2. Scenario S1: coverage lifetime T_{90}, first node dead, coverage integral and readings delivered versus N (mean $\pm$ 95% CI).

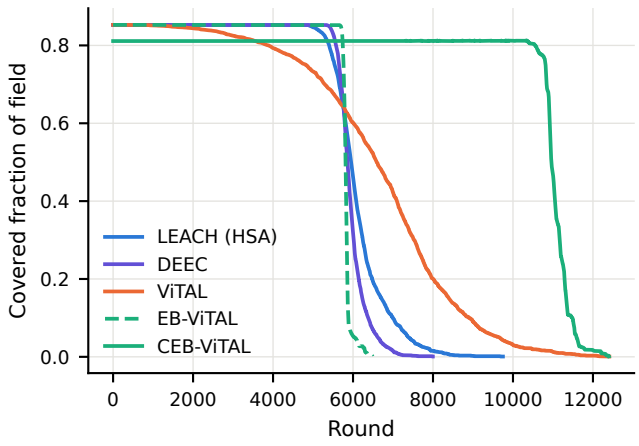


Fig. 3. Covered fraction of the field over time in S1, $N = 300$ (mean of 10 networks).

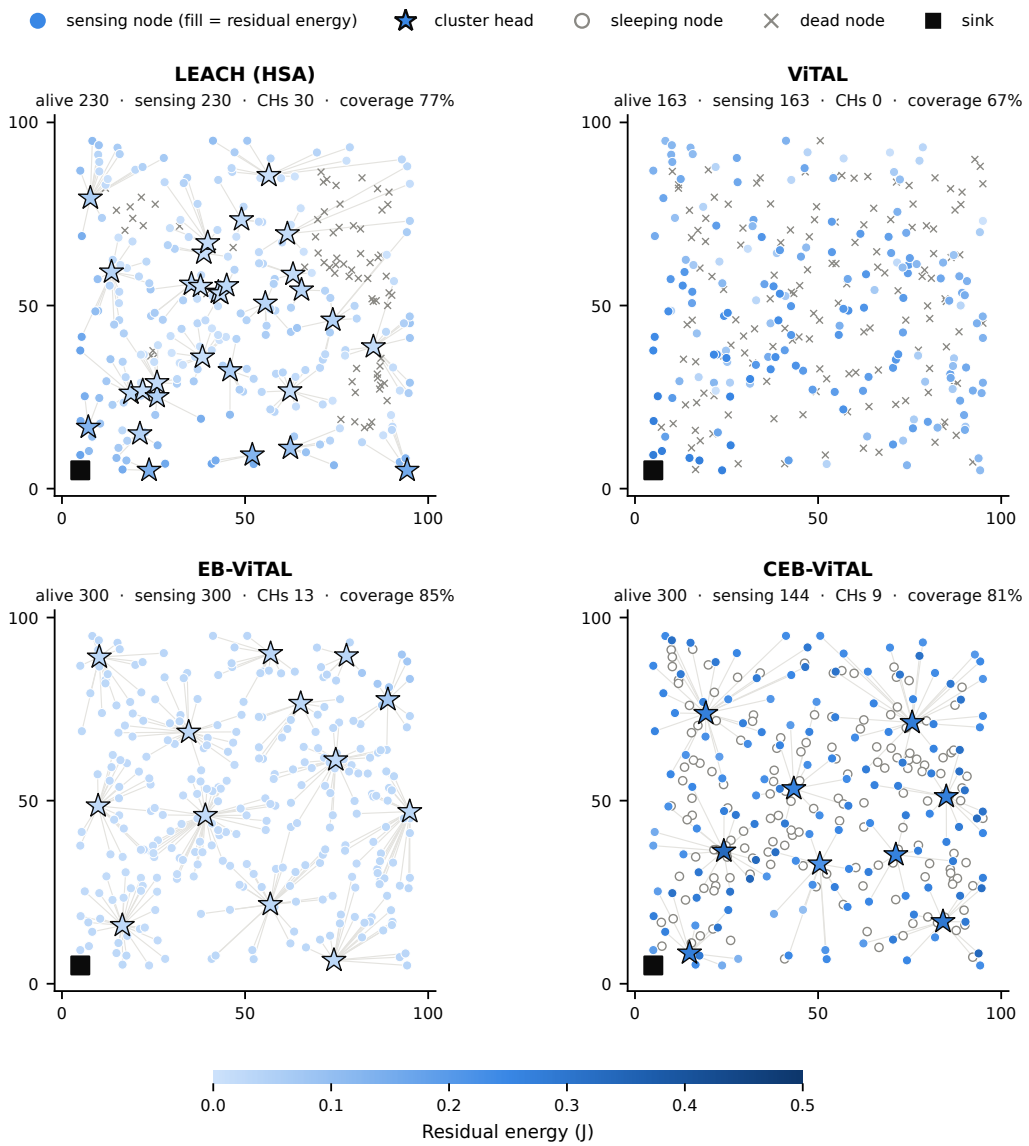


Fig. 4. Network state at round 5500 for one held-out network (S1, $N = 300$). Node fill shows residual energy; stars are CHs, hollow circles are sleeping nodes, crosses are dead nodes, and grey lines join members to their CH.

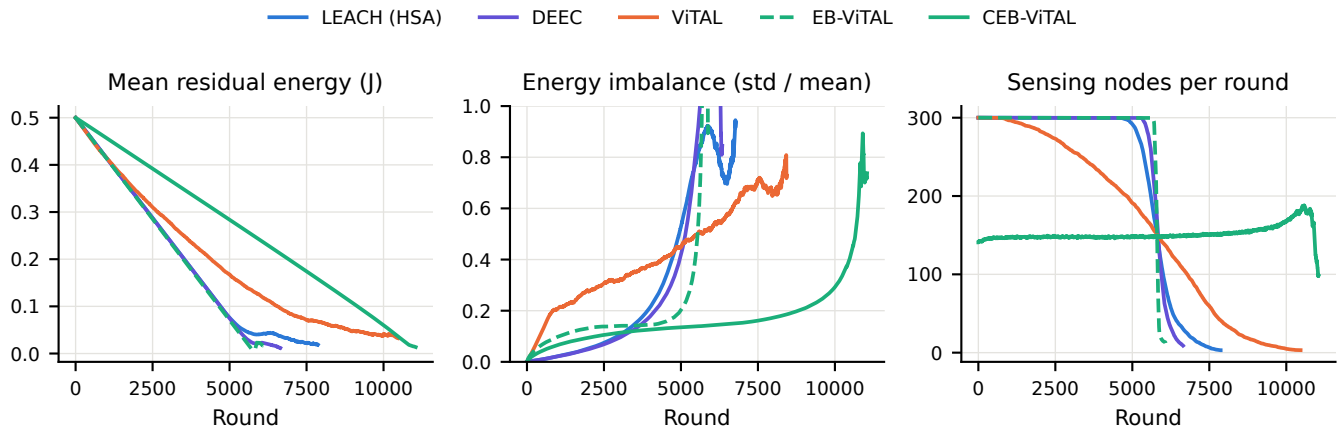


Fig. 5. Network dynamics in S1, $N = 300$ (mean of 10 held-out networks): mean residual energy of alive nodes, energy imbalance (standard deviation over mean, shown while at least 10% of nodes are alive), and number of sensing nodes per round.

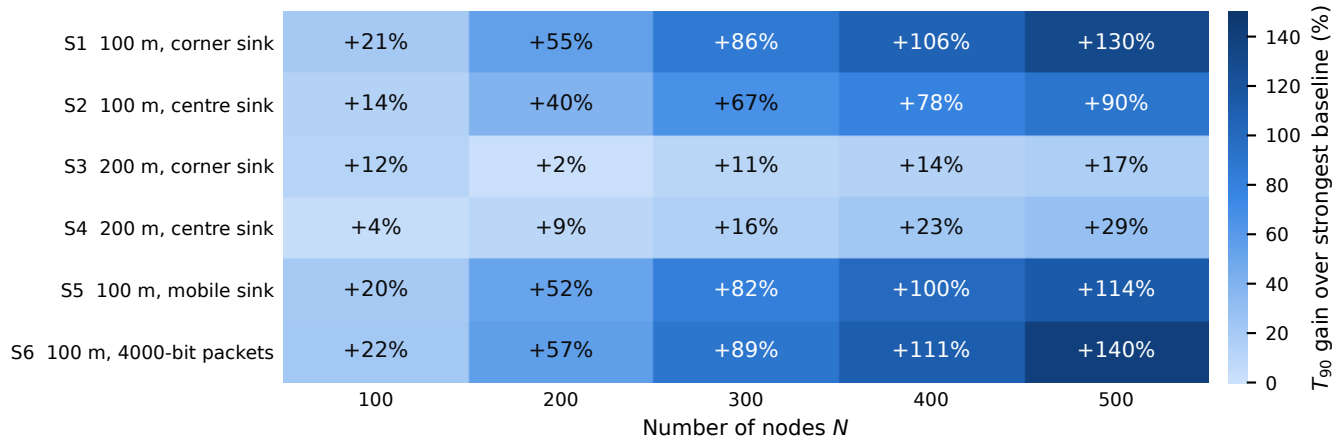


Fig. 6. T_{90} gain of CEB-ViTAL over the strongest always-on protocol for every scenario and network size (mean over 30 held-out networks).

TABLE IV
CEB-ViTAL VERSUS THE STRONGEST ALWAYS-ON PROTOCOL (HIGHEST MEAN T_{90} AMONG LEACH (HSA), DEEC, ViTAL AND EB-ViTAL) IN EVERY SCENARIO. RELATIVE CHANGE IN THE MEAN OVER 30 HELD-OUT NETWORKS.

Scen.	N	Strongest baseline	T_{90}	FND	Cov. int.	LND	Readings
S1	100	EB-ViTAL	+21%	+21%	+15%	+21%	−1%
	300	EB-ViTAL	+86%	+88%	+80%	+89%	−2%
	500	EB-ViTAL	+130%	+134%	+148%	+143%	−3%
S2	100	EB-ViTAL	+14%	+17%	+11%	+7%	−1%
	300	ViTAL	+67%	+696%	+27%	+24%	−15%
	500	ViTAL	+90%	+1561%	+68%	+75%	−16%
S3	100	EB-ViTAL	+12%	+141%	0%	+1%	−1%
	300	EB-ViTAL	+11%	+184%	+9%	+11%	−1%
	500	EB-ViTAL	+17%	+251%	+20%	+18%	−1%
S4	100	EB-ViTAL	+4%	+7%	0%	+5%	−2%
	300	EB-ViTAL	+16%	+18%	+11%	+19%	−3%
	500	EB-ViTAL	+29%	+30%	+24%	+37%	−3%
S5	100	EB-ViTAL	+20%	+20%	+15%	+24%	−1%
	300	DEEC	+82%	+89%	+76%	+93%	−2%
	500	ViTAL	+114%	+2036%	+82%	+66%	−9%
S6	100	EB-ViTAL	+22%	+22%	+16%	+22%	0%
	300	EB-ViTAL	+89%	+91%	+82%	+92%	−1%
	500	EB-ViTAL	+140%	+141%	+148%	+147%	−1%

TABLE V
ABLATION: CHANGE IN CEB-ViTAL RELATIVE TO EACH VARIANT, SHOWN AS T_{90} / FND / COVERAGE INTEGRAL. †THE VARIANT'S MEAN T_{90} IS BELOW 100 ROUNDS, SO A RATIO IS NOT MEANINGFUL.

Scen.	Variant	N=100	N=300	N=500
S1	No CovSched (always on)	+21% / +21% / +15%	+86% / +88% / +80%	+130% / +134% / +148%
	Random sleepers, same count	≈0† / −1% / +10%	≈0† / −9% / +17%	≈0† / −26% / +12%
	LEACH clustering instead of EB	+23% / +23% / +6%	+61% / +61% / +23%	+58% / +59% / +25%
	No multi-hop (M3)	0% / 0% / 0%	0% / 0% / 0%	0% / 0% / 0%
S3	No CovSched (always on)	+12% / +141% / 0%	+11% / +184% / +9%	+17% / +251% / +20%
	Random sleepers, same count	+35% / +129% / +1%	≈0† / +137% / +7%	≈0† / +146% / +10%
	LEACH clustering instead of EB	+41% / +41% / +8%	+110% / +108% / +24%	+140% / +138% / +36%
	No multi-hop (M3)	+3% / +3% / +3%	0% / +1% / +2%	+2% / +4% / +3%

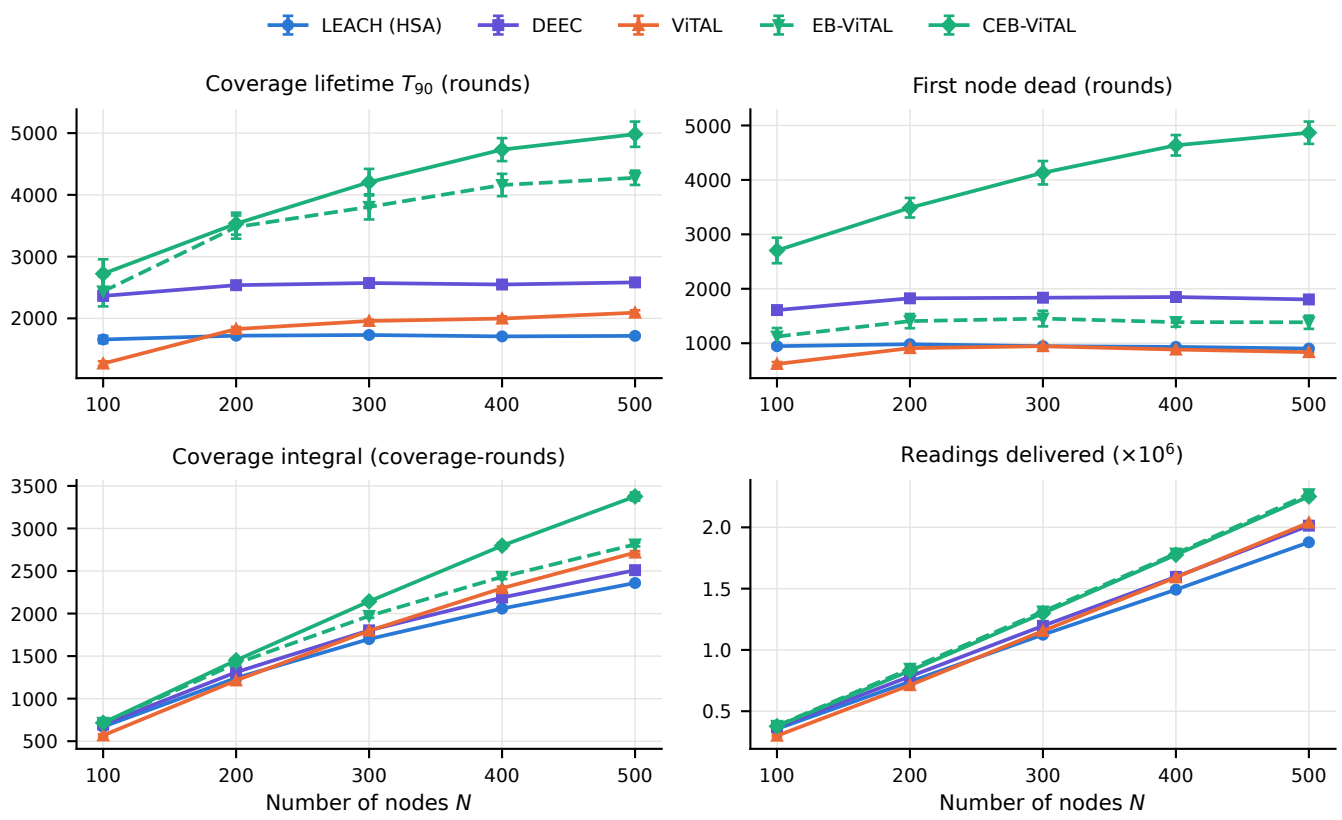


Fig. 7. Scenario S3 (200 m field, sink at (5,5)).

TABLE VI
SLEEP-ENERGY ROBUSTNESS (POST HOC). CEB-VITAL VERSUS THE STRONGEST ALWAYS-ON PROTOCOL, SHOWN AS T_{90} / FND / COVERAGE INTEGRAL.

Scen.	N	$f = 0$	$f = 0.05$	$f = 0.25$	$f = 1$
S1	100	+21% / +21% / +15%	+20% / +21% / +14%	+18% / +18% / +12%	+10% / +10% / +4%
	300	+86% / +88% / +80%	+83% / +86% / +78%	+75% / +75% / +66%	+37% / +35% / +28%
	500	+130% / +134% / +148%	+125% / +128% / +141%	+108% / +105% / +115%	+56% / +49% / +45%
S3	100	+12% / +141% / 0%	+11% / +140% / 0%	+10% / +137% / 0%	+6% / +129% / −1%
	300	+11% / +184% / +9%	+10% / +182% / +9%	+6% / +174% / +7%	−3% / +144% / +3%
	500	+17% / +251% / +20%	+15% / +248% / +20%	+10% / +234% / +17%	0% / +157% / +9%

Data and Code Availability

The simulator, the pre-registration, the raw run data (6,900 test runs and 1,080 robustness runs) and the analysis scripts will be released at [repository URL]. Every number in this paper is generated from those data by the released scripts.

Declaration on the Use of AI Tools

[Authors: state how AI tools were used in coding, analysis and writing, as required by the target journal's policy, and confirm that you have verified all content.]

Author Contributions

[Authors: state each author's contribution, e.g. using the CRediT taxonomy.]

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